Transformations — Wrong Centre, Vector or Scale Factor Answer Key

High School Geometry

Quick Answers

1. 7, -3	3. 5, 5.66, 4	5. 8, 5, 7.21
2. -5, -4	4. 5, 15, 3	6. —

Curriculum

Level: High School Geometry
HSG-CO.A–HSG-CO.D – *problems 1, 2, 3*
HSG-SRT.A.1–A.3, HSG-SRT.B.4–B.5 – *problems 4, 5, 6*
Difficulty 1–5 of 5 across 14 tagged checks.

Common wrong answers

1. *If they got -7: reversed the subtraction, which reverses the whole vector*
1. *If they got 3: reversed the subtraction, which reverses the whole vector*
2. *If they got 5: subtracted image from pre-image, which drops both minus signs and points the vector backwards*
2. *If they got 4: subtracted image from pre-image, which drops both minus signs and points the vector backwards*
3. *If they got 5: applied only the horizontal part of the vector and left the second coordinate unchanged*
4. *If they got 6: subtracted the two x-coordinates and called the difference the scale factor*
4. *If they got 10: subtracted the two distances instead of dividing them*
5. *If they got 10: multiplied the coordinate by the scale factor as if the centre were the origin*
5. *If they got 6: multiplied the coordinate by the scale factor as if the centre were the origin*
5. *If they got 9.43: measured from the origin instead of from the centre of dilation*

1. Translation vector taking $A(-3, 2)$ to $A'(4, -1)$.

Step 1: a translation vector records how far the point moved, so each component is *image minus pre-image.* Write the subtraction in that order before touching any numbers.
Step 2: horizontal component: $4-(-3)$, so $a = \boxed{7}$.
Step 3: vertical component: $-1-2$, so $b = \boxed{-3}$. The translation vector is $\langle 7,\ -3\rangle$: seven right and three down.
Watch for: $\langle -7,\ 3\rangle$. That is the same subtraction done backwards, and it is the vector that maps A' back to A — a real transformation, just not this one.

2. Find and fix: Lee's $P(2,\,5) \to P'(-3,\,1)$.

Step 1: name the error. Lee computed pre-image minus image: $2-(-3)=5$ and $5-1=4$, giving $\langle 5,\ 4\rangle$. Reversing the subtraction reverses the vector, so Lee has described the translation that maps P' back to P.
Step 2: redo it in the right order. Horizontal: $-3-2$, so the first component is $\boxed{-5}$.
Step 3: vertical: $1-5$, so the second component is $\boxed{-4}$. The correct vector is $\langle -5,\ -4\rangle$: five left and four down.
Step 4: sanity check by applying it. $(2+(-5),\ 5+(-4))=(-3,\ 1)$, which is P'. Lee's vector applied to P would have landed at $(7,\ 9)$ — nowhere near the image.
Full-credit answer says: "reversing the order gives the inverse translation."

3. Find and fix: Kim's images of $A(1,\,2)$ and $B(5,\,5)$ under $\langle 3,\ -1\rangle$.

Step 1 (part a): $AB=\sqrt{(5-1)^2+(5-2)^2}=\sqrt{16+9}=\sqrt{25}$, so $AB=\boxed{5}$.
Step 2 (part b): using Kim's coordinates $A'(4,\,1)$ and $B'(8,\,5)$, $A'B' = \sqrt{(8-4)^2+(5-1)^2}=\sqrt{16+16}=\sqrt{32}$, so $A'B'\approx\boxed{5.66}$.
Step 3 (part c): a translation is a rigid motion, so it preserves distance. The image segment cannot be longer than the original, and $5.66\neq 5$, so at least one image point is wrong.
Step 4: apply the vector to B properly. The y-component is -1, so $5+(-1)$ gives the corrected y-coordinate $\boxed{4}$, and $B'(8,\,4)$. Checking: $A'B'=\sqrt{16+9}=5$, which now matches AB.
Watch for: a y-coordinate left at 5. That is Kim's error — the horizontal part of the vector was applied and the vertical part was not.

4. Find and fix: Jo's scale factor for $P(3,\,4) \to P'(9,\,12)$, centre the origin.

Step 1 (part a): the distance from the origin to P is $\sqrt{3^2+4^2}=\sqrt{9+16}=\sqrt{25}$, so $OP=\boxed{5}$.
Step 2 (part b): the distance from the origin to P' is $\sqrt{9^2+12^2}=\sqrt{81+144}=\sqrt{225}$, so $OP'=\boxed{15}$.
Step 3 (part c): the scale factor is the ratio of the image distance to the pre-image distance, $k=\dfrac{15}{5}$, so $k=\boxed{3}$. Every image point sits three times as far from the centre as its pre-image, which is what a scale factor means.
Step 4: why a difference cannot be a scale factor. A scale factor multiplies every distance from the centre by the same number. A difference is not scale invariant — doubling every coordinate of the problem would leave the ratio at 3 but turn Jo's difference into 12 — so a difference cannot describe a dilation at all.
Watch for: Jo's 6, which is $9-3$, and the related error $15-5=10$. Both are subtractions where a division was needed.

5. Find and fix: Ravi's dilation, centre $C(2,\,1)$, factor 2, point $A(5,\,3)$.

Step 1 (part a): doubling the coordinates is the rule for a dilation centred at the *origin*. Ravi's method silently used the origin as the centre instead of C.
Step 2 (part b): use $A'=C+k\,(A-C)$. Horizontally, $2+2(5-2)=2+6$, so the x-coordinate is $\boxed{8}$.
Step 3: vertically, $1+2(3-1)=1+4$, so the y-coordinate is $\boxed{5}$, and $A'(8,\,5)$.
Step 4 (part c): $CA'=\sqrt{(8-2)^2+(5-1)^2}=\sqrt{36+16}=\sqrt{52}$, so $CA'\approx\boxed{7.21}$. Check the factor: $CA=\sqrt{9+4}=\sqrt{13}\approx 3.61$, and $7.21\div 3.61=2$, the scale factor we were given.
Watch for: Ravi's $(10,\ 6)$, which is 2 times each coordinate. Also watch for a distance of about 9.43 — that is the distance from the *origin* to A', measured from the wrong point again.

6. Show what you know: two centres, one scale factor.

Open task — flagged for manual review. A model answer is below; accept any correct sketch and any equivalent explanation.

(a) Model sketch: a triangle, a centre C_1, and the image found by doubling each distance along the ray from C_1; then a second centre C_2 elsewhere, and a second image found the same way from C_2.

(b) Model answer: the two images are the *same size and shape* as each other — both are the original enlarged by a factor of 2, so they are congruent to one another — but they sit in *different places.* A dilation moves each point along the ray from its own centre, so changing the centre changes where every image point lands even though it does not change any length ratio. Equivalently, the two images differ by a translation.

(c) Model answer: draw the line through the point and its image. The centre must lie on that line, because a dilation moves each point along the ray from the centre. Repeat with a second point and its image; the centre is where the two lines meet. It works because the centre is the one point a dilation leaves fixed, so it is the only point common to every such line.

What a full-credit answer must contain: a sketch with both centres and both images labelled; the observation that the images are congruent to each other but differently placed; and a centre-finding method that uses the intersection of at least two point-to-image lines, with a reason.